

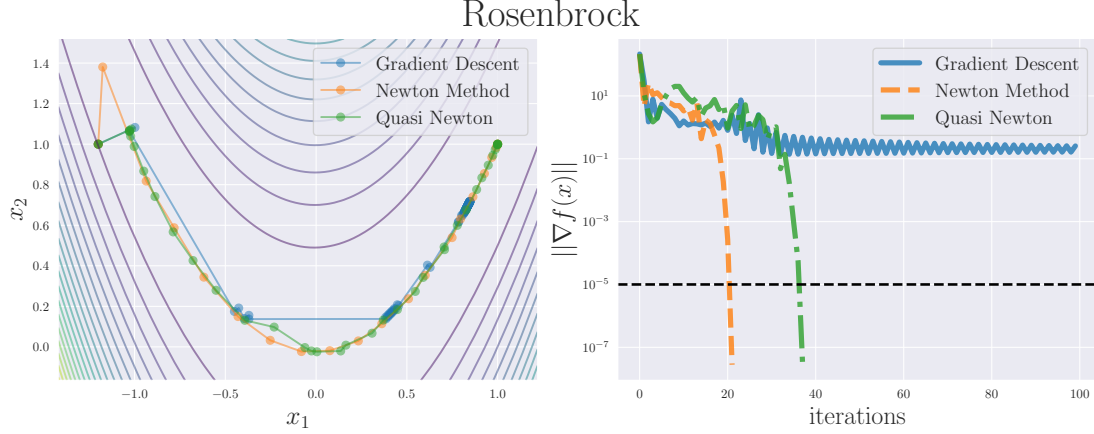


Figure 1: Comparison of Newton's method, quasi-Newton methods, and gradient descent

1 Quasi-Newton Methods

In this section, we discuss quasi-Newton methods. Quasi-Newton methods are based on Newton's method, but aim to reduce its primary drawback, namely, the high computational cost of evaluating the Hessian matrix. Specifically, instead of using the true Hessian matrix $\nabla^2 f(x_k)$, an approximation matrix B_k is employed, thereby achieving fast convergence while keeping the computational cost low.

Figure 1 shows a comparison of Newton's method, quasi-Newton methods, and gradient descent. Gradient descent is a method that simply updates in the direction opposite to the gradient. Although its per-iteration computational cost is the lowest, its convergence is slow. Newton's method converges in the fewest iterations, but has the highest computational cost per step. Quasi-Newton methods lie between these two extremes, striking a balance between them. In particular, when comparing methods in terms of computation time rather than the number of iterations, quasi-Newton methods often exhibit the best performance.

Line-search-based quasi-Newton methods generate a sequence $\{x_k\}_{k=0}^{\infty}$ converging to the optimal solution x^* as follows:

$$x_{k+1} = x_k - \alpha_k B_k^{-1} \nabla f(x_k) = x_k - \alpha_k H_k \nabla f(x_k)$$

Here, $\alpha_k > 0$ is the step size determined by line search, B_k is an approximation of the Hessian matrix $\nabla^2 f(x_k)$ at the point x_k , and $H_k := B_k^{-1}$ denotes its inverse.

The core of quasi-Newton methods lies in how to update B_k (or its inverse H_k) at each iteration so that it approaches the Hessian matrix $\nabla^2 f(x_k)$. Figure 2 illustrates this concept. First, around the current point x_k , a quadratic approximation model of the objective function f is constructed using B_k . Next, this quadratic model is minimized to

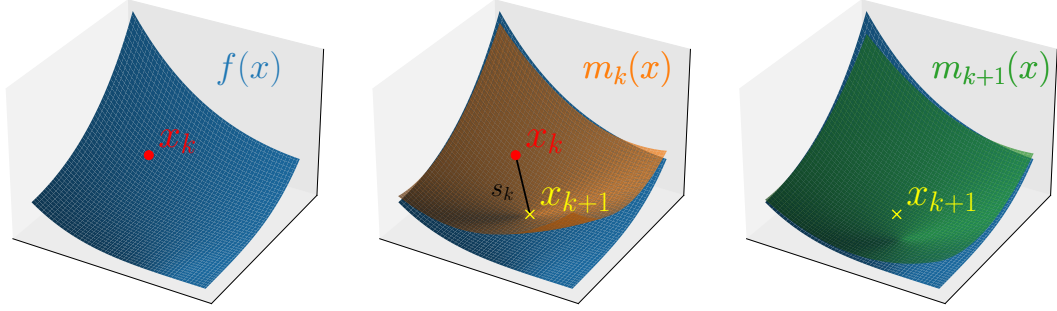


Figure 2: Conceptual illustration of quasi-Newton methods. (1) The objective function f (blue surface) and the current point x_k (red dot). (2) The quadratic model induced by the current Hessian approximation (orange surface) and its minimizer x_{k+1} (yellow cross). (3) The updated quadratic model based on the new point x_{k+1} (green surface).

obtain the next point x_{k+1} . After obtaining x_{k+1} , the approximation matrix B_k is updated to B_{k+1} using the gradient information at x_k and x_{k+1} . This procedure is repeated until convergence, which constitutes the quasi-Newton method.

In what follows, we introduce the secant condition that the approximate Hessian matrix B_k is required to satisfy, and present several representative update formulas for such B_k and H_k .

1.1 Secant Condition

In this section as well, let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be of class C^2 . Suppose that a symmetric matrix B_k is given as an approximation of the Hessian matrix $\nabla^2 f(x_k)$. We determine an approximation Hessian B_{k+1} at the next point x_{k+1} . Although there are infinitely many possible candidates for such a matrix, it is natural to impose symmetry on B_{k+1} as well, given that the true Hessian matrix is symmetric. Define the step and gradient difference as

$$s_k = x_{k+1} - x_k, \quad y_k = \nabla f(x_{k+1}) - \nabla f(x_k),$$

where we assume that $y_k^\top s_k \neq 0$ and $s_k \neq 0$. Here, the s in s_k stands for step. Using the Taylor expansion of $\nabla f(x_k)$, we have

$$\begin{aligned} \nabla f(x_k) &= \nabla f(x_{k+1}) + \nabla^2 f(x_{k+1})(x_k - x_{k+1}) + \mathcal{O}(\|x_k - x_{k+1}\|^2) \\ &\approx \nabla f(x_{k+1}) + \nabla^2 f(x_{k+1})(x_k - x_{k+1}) \\ &\approx \nabla f(x_{k+1}) + B_{k+1}(x_k - x_{k+1}) \end{aligned}$$

Requiring the above approximation to hold as an equality yields

$$B_{k+1}(x_{k+1} - x_k) = \nabla f(x_{k+1}) - \nabla f(x_k),$$

or equivalently,

$$B_{k+1}s_k = y_k. \quad (1)$$

The relation in eq. (1) is called the secant condition, or the quasi-Newton equation.

1.2 Representative Quasi-Newton Update Rules

Given B_k , s_k , and y_k , there exist many update rules that produce B_{k+1} satisfying the secant condition. We introduce several representative ones with their derivations [2]. In this subsection only, for brevity, we abbreviate B_k , B_{k+1} , s_k , and y_k as B , $\bar{B}$, s , and y , respectively.

1.2.1 Broyden's Update

Broyden's update is one of the most fundamental quasi-Newton update formulas, but it is not popular in practice due to its lack of symmetry preservation. The update formulas are given by

$$\bar{B}_{\text{Broyden}} = B + \frac{(y - Bs)s^\top}{s^\top s}, \quad (2)$$

$$\bar{H}_{\text{Broyden}} = H + \frac{s - Hy}{s^\top Hy} s^\top H. \quad (3)$$

Derivation We derive this formula from simple structural assumptions [2, Section 4].

Proposition 1. *Assume that $\bar{B}$ satisfies the secant condition*

$$\bar{B}s = y, \quad (4)$$

and the action constraint

$$\bar{B}z = Bz \quad \text{for all } z \in \mathbb{R}^n \text{ such that } z^\top s = 0. \quad (5)$$

Then $\bar{B}$ is uniquely determined and it is $\bar{B}_{\text{Broyden}}$ defined by eq. (2).

Proof. A basis for $\mathbb{R}^n$ can be constructed from s and a basis for the orthogonal complement of s . Since eqs. (4) and (5) completely determine the action of $\bar{B}$ with respect to this basis, $\bar{B}$ is uniquely determined.

We now show that $\bar{B}_{\text{Broyden}}$ defined by eq. (2) satisfies eqs. (4) and (5). Let z be any vector satisfying $z^\top s = 0$. Then

$$\bar{B}_{\text{Broyden}}s = \left(B + \frac{(y - Bs)s^\top}{s^\top s} \right) s = Bs + (y - Bs) = y,$$

$$\bar{B}_{\text{Broyden}} z = \left(B + \frac{(y - Bs)s^\top}{s^\top s} \right) z = Bz + (z^\top s) \frac{y - Bs}{s^\top s} = Bz.$$

Hence $\bar{B}_{\text{Broyden}}$ indeed satisfies eqs. (4) and (5). Therefore, by uniqueness, we have $\bar{B} = \bar{B}_{\text{Broyden}}$. $\square$

Broyden's update can also be characterized as a minimal-change update in the Frobenius norm.

Proposition 2 ([2], Theorem 4.1). *Let $B \in \mathbb{R}^{n \times n}$, $y \in \mathbb{R}^n$, and $s \in \mathbb{R}^n \setminus \{0\}$ be given. Then the matrix $\bar{B}$ defined by eq. (2) is the unique solution of*

$$\begin{aligned} & \underset{\tilde{B} \in \mathbb{R}^{n \times n}}{\text{minimize}} && \|\tilde{B} - B\|_F \\ & \text{subject to} && \tilde{B}s = y. \end{aligned}$$

Proof. The function $\tilde{B} \mapsto \|\tilde{B} - B\|_F$ is strictly convex on $\mathbb{R}^{n \times n}$. The constraint set

$$\{\tilde{B} \in \mathbb{R}^{n \times n} : \tilde{B}s = y\} \quad (6)$$

is affine and hence convex. Therefore, the optimization problem admits at most one minimizer.

To show that $\bar{B} = \bar{B}_{\text{Broyden}}$ defined by eq. (2) is indeed the minimizer, we compute:

$$\|\bar{B}_{\text{Broyden}} - B\|_F^2 = \left\| \frac{(y - Bs)s^\top}{s^\top s} \right\|_F^2 = \left\| (\bar{B} - B) \frac{ss^\top}{s^\top s} \right\|_F^2 \leq \|\bar{B} - B\|_F^2,$$

where we used the sub-multiplicativity of Frobenius norm and the fact that $\|ss^\top / (s^\top s)\|_F = 1$ in the last inequality. Thus, $\bar{B} = \bar{B}_{\text{Broyden}}$. $\square$

Broyden's update is characterized by two properties: it satisfies the secant condition and preserves the action on vectors orthogonal to s . Additionally, it is the unique minimal-change update in the Frobenius norm subject to the secant constraint.

1.2.2 SR1 Update

The Symmetric Rank-One (SR1) update [7] is a fundamental quasi-Newton method that maintains symmetry throughout the update process. The update formulas are given by

$$\bar{B}_{\text{SR1}} = B + \frac{(y - Bs)(y - Bs)^\top}{(y - Bs)^\top s}, \quad (7)$$

$$\bar{H}_{\text{SR1}} = H + \frac{(s - Hy)(s - Hy)^\top}{(s - Hy)^\top y}. \quad (8)$$

Derivation To derive eq. (7), we construct the updated matrix $\bar{B}$ as a rank-one update. That is, we assume that for some vector $z \in \mathbb{R}^n$,

$$\bar{B}_{\text{SR1}} = B + zz^\top. \quad (9)$$

For this update to satisfy the secant condition $\bar{B}_{\text{SR1}}s = y$, assuming $z^\top s \neq 0$, we require

$$Bs + zz^\top s = y, \quad (10)$$

which yields

$$z = \frac{y - Bs}{z^\top s}. \quad (11)$$

To determine $z^\top s$, we take the inner product of both sides of eq. (11) with s :

$$z^\top s = \frac{(y - Bs)^\top s}{z^\top s}. \quad (12)$$

Rearranging eq. (12) yields the key relation

$$(z^\top s)^2 = (y - Bs)^\top s. \quad (13)$$

Substituting eq. (13) into eq. (11) and using eq. (9), we obtain

$$\bar{B}_{\text{SR1}} = B + zz^\top \quad (14)$$

$$= B + \frac{(y - Bs)(y - Bs)^\top}{(z^\top s)^2} \quad (15)$$

$$= B + \frac{(y - Bs)(y - Bs)^\top}{(y - Bs)^\top s}, \quad (16)$$

which is the SR1 update formula stated in eq. (7).

Remarks The SR1 update requires $(y - Bs)^\top s \neq 0$ to be well-defined. When $(y - Bs)^\top s = 0$, the denominator in eq. (7) vanishes, and the update is typically skipped in practice. This situation can occur when the secant condition cannot be satisfied by a symmetric rank-one update, indicating that more sophisticated update strategies are needed.

1.2.3 Powell Symmetric Broyden (PSB) Update

The Powell Symmetric Broyden (PSB) update [3] is one of the most important quasi-Newton update rules. The update formulas are given by

$$\bar{B}_{\text{PSB}} = B + \frac{(y - Bs)s^\top + s(y - Bs)^\top}{s^\top s} - \frac{s^\top (y - Bs)}{(s^\top s)^2} ss^\top, \quad (17)$$

$$\bar{H}_{\text{PSB}} = H + \frac{(s - Hy)y^\top + y(s - Hy)^\top}{y^\top y} - \frac{y^\top (s - Hy)}{(y^\top y)^2} yy^\top. \quad (18)$$

Derivation To understand the motivation behind this formula, we note that in the SR1 update, the rank-one update was formulated symmetrically as $\bar{B}_{\text{SR1}} = B + zz^\top$. We may relax the requirement of maintaining symmetry throughout the update process. Instead, we can consider an asymmetric rank-one update of the form $zc^\top$, where the final result is symmetrized afterward.

For a given vector $c \in \mathbb{R}^n$ with $c^\top s \neq 0$, we define

$$z = \frac{y - Bs}{c^\top s} \quad (19)$$

and perform the asymmetric update

$$C_1 := B + \frac{(y - Bs)c^\top}{c^\top s}. \quad (20)$$

Since C_1 is generally not symmetric, we symmetrize it via

$$C_2 = \frac{C_1 + C_1^\top}{2}. \quad (21)$$

However, the symmetrized matrix C_2 may not satisfy the secant condition $C_2 s = y$. Therefore, we iterate this process:

$$\begin{cases} C_0 = B \\ C_{2t+1} = C_{2t} + \frac{(y - C_{2t}s)c^\top}{c^\top s} & \text{(asymmetric update)} \\ C_{2t+2} = \frac{C_{2t+1} + C_{2t+1}^\top}{2} & \text{(symmetrization)} \end{cases} \quad (22)$$

The key result is that the sequence $\{C_{2t}\}_{t=0}^\infty$ converges to a symmetric matrix satisfying the secant condition, as formalized in the following proposition.

Proposition 3 ([2], Lemma 7.2). *The sequence $\{C_{2t}\}_{t=0}^\infty$ defined by eq. (22) converges, and the limit is given by*

$$\lim_{t \rightarrow \infty} C_{2t} = B + \frac{(y - Bs)c^\top + c(y - Bs)^\top}{c^\top s} - \frac{(y - Bs)^\top s}{(c^\top s)^2} cc^\top. \quad (23)$$

Proof. We first analyze the even subsequence. Define

$$G_k := C_{2k}$$

for $k = 0, 1, 2, \dots$. By construction, each G_k is symmetric. From eq. (22) and the definition, we have

$$G_{k+1} = G_k + \frac{1}{2c^\top s} \left((y - G_k s)c^\top + c(y - G_k s)^\top \right).$$

Let us introduce the error vector

$$w_k := y - G_k s \quad (24)$$

and then we have

$$G_{k+1} = G_k + \frac{1}{2c^\top s} (w_k c^\top + c w_k^\top). \quad (25)$$

Substituting eq. (25) into eq. (24) yields

$$w_{k+1} = y - \left(G_k + \frac{1}{2c^\top s} (w_k c^\top + c w_k^\top) \right) s \quad (26)$$

$$= w_k - \frac{1}{2} w_k - \frac{w_k^\top s}{2c^\top s} c \quad (27)$$

$$= \frac{1}{2} \left(w_k - \frac{w_k^\top s}{c^\top s} c \right). \quad (28)$$

Hence

$$w_{k+1} = P w_k, \quad P := \frac{1}{2} \left(I - \frac{c s^\top}{c^\top s} \right). \quad (29)$$

The matrix $c s^\top / c^\top s$ has rank one and eigenvalues $1, 0, \dots, 0$. Therefore, P has one eigenvalue 0 and all remaining eigenvalues equal to $1/2$. In particular, its spectral radius is $1/2 < 1$. Thus, the Neumann series converges and

$$\sum_{k=0}^{\infty} w_k = \sum_{k=0}^{\infty} P^k (y - B s) \quad (\text{from } w_0 = y - B s) \quad (30)$$

$$= (I - P)^{-1} (y - B s) \quad (31)$$

$$= 2 \left(I - \frac{1}{2} \frac{c s^\top}{c^\top s} \right) (y - B s). \quad (\text{from the definition of } P) \quad (32)$$

Note that the last equation follows from

$$2(I - P) \left(I - \frac{1}{2} \frac{c s^\top}{c^\top s} \right) = \left(I + \frac{c s^\top}{c^\top s} \right) \left(I - \frac{1}{2} \frac{c s^\top}{c^\top s} \right) = I.$$

In particular, $\|w_k\| \rightarrow 0$ as $k \rightarrow \infty$. Hence, from eq. (25), we have

$$\lim_{k \rightarrow \infty} G_k = B + \frac{1}{2c^\top s} \sum_{k=0}^{\infty} (w_k c^\top + c w_k^\top) \quad (33)$$

$$= B + \left(\sum_{k=0}^{\infty} w_k \right) \frac{c^\top}{2c^\top s} + \frac{c}{2c^\top s} \left(\sum_{k=0}^{\infty} w_k \right)^\top \quad (34)$$

$$= B + \frac{(y - Bs)c^\top + c(y - Bs)^\top}{c^\top s} - \frac{(y - Bs)^\top s}{(c^\top s)^2} cc^\top. \quad (35)$$

Thus $G_k \rightarrow \bar{B}$. Finally,

$$C_{2k+1} = G_k + \frac{w_k c^\top}{c^\top s}. \quad (36)$$

Since $G_k \rightarrow \bar{B}$ and $\|w_k\| \rightarrow 0$, we obtain

$$C_{2k+1} - G_k \rightarrow 0. \quad (37)$$

Therefore both subsequences $\{C_{2k}\}$ and $\{C_{2k+1}\}$ converge to $\bar{B}$, and hence

$$C_k \rightarrow \bar{B}. \quad (38)$$

This completes the proof. $\square$

When $c = s$, the general formula in eq. (23) simplifies to the standard PSB update formula:

$$\bar{B}_{\text{PSB}} = B + \frac{(y - Bs)s^\top + s(y - Bs)^\top}{s^\top s} - \frac{(y - Bs)^\top s}{(s^\top s)^2} ss^\top, \quad (39)$$

which matches eq. (17).

Remarks The choice of $c = s$ is motivated by ensuring positive definiteness of the resulting update. The PSB update can be viewed as the limit of an iterative symmetrization process applied to asymmetric rank-one updates.

1.2.4 DFP Update

The Davidon–Fletcher–Powell (DFP) update [7] is a classical quasi-Newton update formula. The update formulas are given by

$$\bar{B}_{\text{DFP}} = \left(I - \frac{ys^\top}{y^\top s}\right) B \left(I - \frac{sy^\top}{y^\top s}\right) + \frac{yy^\top}{y^\top s}, \quad (40)$$

$$\bar{H}_{\text{DFP}} = H - \frac{Hyy^\top H}{y^\top Hy} + \frac{ss^\top}{y^\top s}. \quad (41)$$

Derivation In the PSB update rule discussed earlier, we substituted $c = s$, but we can consider taking a different c . Specifically, we consider choosing c such that B_{k+1} becomes positive definite. Substituting $c = y$ yields the alternative form:

$$\bar{B}_{\text{DFP}} = B + \frac{(y - Bs)y^\top + y(y - Bs)^\top}{y^\top s} - \frac{(y - Bs)^\top s}{(y^\top s)^2} yy^\top. \quad (42)$$

1.2.5 BFGS Update

The Broyden–Fletcher–Goldfarb–Shanno (BFGS) update is one of the most widely used quasi-Newton methods. The update formulas are given by

$$\bar{B}_{\text{BFGS}} = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}, \quad (43)$$

$$\bar{H}_{\text{BFGS}} = \left(I - \frac{sy^\top}{y^\top s}\right) H \left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s}. \quad (44)$$

Derivation This update can be derived by considering the dual of the DFP update. Specifically, we seek an update that minimizes the change in the inverse Hessian approximation H while satisfying the secant condition. Further details are provided in the following subsection.

1.3 BFGS Method

In this subsection, we focus on the BFGS update and provide a detailed derivation of its formula. BFGS update is known as one of the most successful quasi-Newton update rules in practice. Recall that the BFGS update formula is given by

$$B_{k+1} = B_k - \frac{B_k s_k s_k^\top B_k}{s_k^\top B_k s_k} + \frac{y_k y_k^\top}{y_k^\top s_k}. \quad (45)$$

1.3.1 Formula for H_k

The inverse of the BFGS update $H_k := B_k^{-1}$ is given by

$$H_{k+1} = \left(I - \frac{s_k y_k^\top}{y_k^\top s_k}\right) H_k \left(I - \frac{y_k s_k^\top}{y_k^\top s_k}\right) + \frac{s_k s_k^\top}{y_k^\top s_k}. \quad (46)$$

We now prove that eq. (46) indeed gives the inverse of the BFGS update.

Proposition 4. Equation (46) gives the exact inverse of the BFGS update, i.e., $H_{k+1} = B_{k+1}^{-1}$.

Proof. The BFGS update can be rewritten in a compact rank-two form

$$B_{k+1} = B_k + UCV^\top \quad (47)$$

where

$$U = [B_k s_k \quad y_k], \quad C = \begin{pmatrix} -\frac{1}{s_k^\top B_k s_k} & 0 \\ 0 & \frac{1}{y_k^\top s_k} \end{pmatrix}, \quad V = [B_k s_k \quad y_k]$$

since

$$UCV^\top = \begin{bmatrix} -\frac{B_k s_k}{s_k^\top B_k s_k} & \frac{y_k}{y_k^\top s_k} \end{bmatrix} \begin{bmatrix} s_k^\top B_k \\ y_k^\top \end{bmatrix} = -\frac{B_k s_k s_k^\top B_k}{s_k^\top B_k s_k} + \frac{y_k y_k^\top}{y_k^\top s_k}. \quad (48)$$

By the Sherman–Morrison–Woodbury identity, we have

$$\begin{aligned} H_{k+1} &= (B_k + UCV^\top)^{-1} \\ &= B_k^{-1} - B_k^{-1}U \left(C^{-1} + V^\top B_k^{-1}U \right)^{-1} V^\top B_k^{-1} \\ &= H_k - \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \left(\begin{pmatrix} -s_k^\top B_k s_k & 0 \\ 0 & y_k^\top s_k \end{pmatrix} + \begin{pmatrix} s_k^\top B_k s_k & s_k^\top y_k \\ y_k^\top s_k & y_k^\top H_k y_k \end{pmatrix} \right)^{-1} \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\ &= H_k - \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \begin{pmatrix} 0 & y_k^\top s_k \\ y_k^\top s_k & y_k^\top H_k y_k + y_k^\top s_k \end{pmatrix}^{-1} \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\ &= H_k - \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \left(-\frac{1}{(y_k^\top s_k)^2} \begin{pmatrix} y_k^\top H_k y_k + y_k^\top s_k & -y_k^\top s_k \\ -y_k^\top s_k & 0 \end{pmatrix} \right) \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\ &= H_k + \frac{1}{(y_k^\top s_k)^2} \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \begin{pmatrix} y_k^\top H_k y_k + y_k^\top s_k & -y_k^\top s_k \\ -y_k^\top s_k & 0 \end{pmatrix} \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\ &= H_k + \frac{1}{(y_k^\top s_k)^2} \left((y_k^\top H_k y_k + y_k^\top s_k) s_k s_k^\top - (y_k^\top s_k) (s_k y_k^\top H_k + H_k y_k s_k^\top) \right) \\ &= \left(I - \frac{s_k y_k^\top}{y_k^\top s_k} \right) H_k \left(I - \frac{y_k s_k^\top}{y_k^\top s_k} \right) + \frac{s_k s_k^\top}{y_k^\top s_k}. \end{aligned}$$

which completes the proof. $\square$

1.3.2 Positive Definiteness of B_{k+1}

An important property of the BFGS update is that if the current approximation B_k is positive definite and the curvature condition $y_k^\top s_k > 0$ holds, then the updated approximation B_{k+1} is also guaranteed to be positive definite.

Proposition 5. *If B_k is positive definite and $y_k^\top s_k > 0$, then B_{k+1} defined by eq. (45) is also positive definite.*

Proof. By assumption, B_k and its inverse H_k are positive definite. For any nonzero vector $v \in \mathbb{R}^n$, we have

$$\begin{aligned} v^\top H_{k+1} v &= v^\top \left(I - \frac{s_k y_k^\top}{y_k^\top s_k} \right) H_k \left(I - \frac{y_k s_k^\top}{y_k^\top s_k} \right) v + v^\top \frac{s_k s_k^\top}{y_k^\top s_k} v \\ &\geq 0 + \frac{(s_k^\top v)^2}{y_k^\top s_k} > 0, \end{aligned}$$

where the first term is nonnegative since H_k is positive definite, and the second term is positive due to the curvature condition $y_k^\top s_k > 0$. Thus, H_{k+1} is positive definite, and consequently, $B_{k+1} = H_{k+1}^{-1}$ is also positive definite. $\square$

1.3.3 KL

We next present a variational characterization of the BFGS update via the Kullback–Leibler (KL) divergence. For zero-mean multivariate Gaussians $\mathcal{N}(0, A^{-1})$ and $\mathcal{N}(0, B^{-1})$, the expression

$$\psi(A) = \text{tr}(A) - \log \det(A)$$

coincides with the KL divergence up to the additive constant $-n$, so minimizing ψ is equivalent to minimizing the KL distance. Then, the BFGS update is given by the solution to the optimization problem

$$\begin{aligned} & \underset{B \in \text{PD}(n)}{\text{minimize}} && \psi(B_k^{-1/2} B B_k^{-1/2}) \\ & \text{subject to} && B s_k = y_k \end{aligned}$$

The constraint $B s_k = y_k$ in the optimization problem is precisely the secant condition discussed earlier, ensuring that the updated matrix interpolates the latest curvature pair.

See literature for proof of these formulations [4, 5]

1.3.4 Trace and Determinant Formulas for the BFGS Update

We also present trace and determinant formulas for the BFGS update, which are useful in analyzing the eigenvalue behavior of the updated matrix. For Hermitian matrices, these quantities correspond to the sum and product of eigenvalues, respectively. Hence, if both the trace and the determinant are appropriately bounded, one may expect the eigenvalues themselves to remain bounded (provided all eigenvalues are positive, for instance). This is closely related to assumptions on the Hessian eigenvalues of the objective function, such as μ -strong convexity and L -smoothness.

Trace Formula The trace of the BFGS-updated matrix has an explicit formula.

Proposition 6 ([7, (6.44)]). *Let $B_+ = B - \frac{B s s^\top B}{s^\top B s} + \frac{y y^\top}{y^\top s}$ be the BFGS update. Then*

$$\text{tr}(B_+) = \text{tr}(B) - \frac{\|B s\|^2}{s^\top B s} + \frac{\|y\|^2}{y^\top s}. \quad (49)$$

Proof. Applying the trace to the BFGS update formula, we have

$$\text{tr}(B_+) = \text{tr}(B) - \text{tr}\left(\frac{B s s^\top B}{s^\top B s}\right) + \text{tr}\left(\frac{y y^\top}{y^\top s}\right). \quad (50)$$

For the second term in eq. (50), note that

$$\text{tr}\left(\frac{Bss^\top B}{s^\top Bs}\right) = \frac{1}{s^\top Bs} \text{tr}\left((Bs)(Bs)^\top\right) = \frac{\|Bs\|^2}{s^\top Bs}. \quad (51)$$

For the third term in eq. (50), we have

$$\text{tr}\left(\frac{yy^\top}{y^\top s}\right) = \frac{1}{y^\top s} \text{tr}(yy^\top) = \frac{\|y\|^2}{y^\top s}. \quad (52)$$

Substituting eqs. (51) and (52) into eq. (50) yields eq. (49). $\square$

Determinant Formula The determinant of the BFGS-updated matrix also admits a closed-form expression.

Proposition 7 ([7, (6.45)]). *Let $B_+ = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}$ be the BFGS update and assume that B is nonsingular. Then*

$$\det(B_+) = \det(B) \frac{y^\top s}{s^\top Bs}. \quad (53)$$

Proof. Recall the rank-two formulation of the BFGS update defined in eq. (47). By the matrix determinant lemma, the matrices U, C, V in eq. (47) satisfy

$$\det(B_{k+1}) = \det(B_k + UCV^\top) = \det(B_k) \det(C) \det(C^{-1} + V^\top B_k^{-1} U), \quad (54)$$

where I_2 is the 2×2 identity matrix. Since $U = V = [B_k s_k \quad y_k]$, we have

$$V^\top B_k^{-1} U = \begin{bmatrix} s_k^\top B_k s_k & s_k^\top y_k \\ y_k^\top s_k & y_k^\top B_k^{-1} y_k \end{bmatrix}$$

and thus

$$C^{-1} + V^\top B_k^{-1} U = \begin{bmatrix} -s_k^\top B_k s_k & 0 \\ 0 & y_k^\top s_k \end{bmatrix} + \begin{bmatrix} s_k^\top B_k s_k & s_k^\top y_k \\ y_k^\top s_k & y_k^\top B_k^{-1} y_k \end{bmatrix} = \begin{bmatrix} 0 & s_k^\top y_k \\ y_k^\top s_k & y_k^\top B_k^{-1} y_k \end{bmatrix}. \quad (55)$$

Thus, combining eqs. (54) and (55), we have

$$\det(B_{k+1}) = \det(B_k) \left(-\frac{1}{s_k^\top B_k s_k} \cdot \frac{1}{y_k^\top s_k} \right) \left(-(s_k^\top y_k)(y_k^\top s_k) \right) = \det(B_k) \frac{y_k^\top s_k}{s_k^\top B_k s_k},$$

which completes the proof. $\square$

1.4 BFGS vs DFP

Although BFGS and DFP are quite symmetric in structure, they exhibit remarkably different practical efficiency when applied to real optimization problems. Powell’s analysis [8] investigates this asymmetry by studying the behavior of both methods on a simple two-dimensional quadratic function. While asymptotic convergence theory often suggests that both methods should perform similarly, Powell reveals that their practical efficiency can differ dramatically, especially when the approximate Hessian is far from the true Hessian.

1.4.1 Problem Setup

Following Powell’s framework, we consider the quadratic function

$$f(x, y) = \frac{1}{2}(x^2 + y^2), \quad (56)$$

with both BFGS and DFP methods using a fixed step size of $\alpha_k = 1$ at every iteration. This choice of unit step size is practical for quadratic functions, where it often satisfies standard line search criteria.

The initial Hessian approximation B_0 is chosen to have eigenvalues 1 and λ_1 , where λ_1 represents the degree of error in the initial approximation. The initial point x_0 is selected as

$$\theta = \arctan\left(\sqrt{\lambda_1}\right), \quad x_0 = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}, \quad (57)$$

which aligns with Powell’s original analysis. See [8] for details on this choice.

The iteration continues until the norm of the current point falls below a tolerance relative to its initial norm. For each value of λ_1 , we record the number of iterations required for convergence.

1.4.2 Experimental Results

The numerical results are presented in Table 1, which partially reproduces the content of Powell’s original tables. The convergence behavior depends critically on the initial eigenvalue λ_1 :

Figure 3 and Figure 4 illustrate the iterative trajectories for specific values of λ_1 . These figures visualize how the two methods navigate toward the minimum (at the origin) from the same initial point, clearly showing the difference in convergence speed and path.

λ_1	BFGS	DFP
0.001	4	3
0.01	5	3
0.1	6	4
1	1	1
10	8	16
100	10	107
1000	12	1006
10000	15	9987

Table 1: Convergence comparison between BFGS and DFP methods for different initial eigenvalues λ_1

1.4.3 Analysis and Discussion

The numerical results reveal a striking asymmetry between BFGS and DFP. When $\lambda_1 > 1$ (i.e., the initial Hessian approximation overestimates the true Hessian curvature), BFGS demonstrates significantly better efficiency. Conversely, when $\lambda_1 < 1$ (i.e., the initial Hessian approximation underestimates the true curvature), DFP performs slightly better than BFGS, though the difference is modest. This trend reversal is predicted by the theoretical symmetry between the two methods, but the magnitude of the difference is notable.

Asymmetry in Hessian Correction The asymmetry of the performance arises from a fundamental difference in how these two methods correct erroneous eigenvalues. The core insight is that correcting erroneously large eigenvalues is more critical than correcting erroneously small ones.

When a Hessian eigenvalue is overestimated, the algorithm takes steps that are too conservative, resulting in slow progress toward the minimum. Correcting such errors requires the update formula to reduce these large eigenvalues toward one. The BFGS update is highly effective at this task.

In contrast, when a Hessian eigenvalue is underestimated, the algorithm takes steps that are slightly too aggressive, but the error is self-correcting. Subsequent gradient computations provide information that helps refine the approximation. Thus, correcting underestimated eigenvalues is inherently easier and requires fewer iterations.

The DFP update struggles with large eigenvalues. In the worst case, it can reduce a large eigenvalue by only a small amount per iteration, potentially requiring as many iterations as the magnitude of the eigenvalue itself. This explains why DFP’s performance degrades catastrophically as λ_1 increases beyond one.

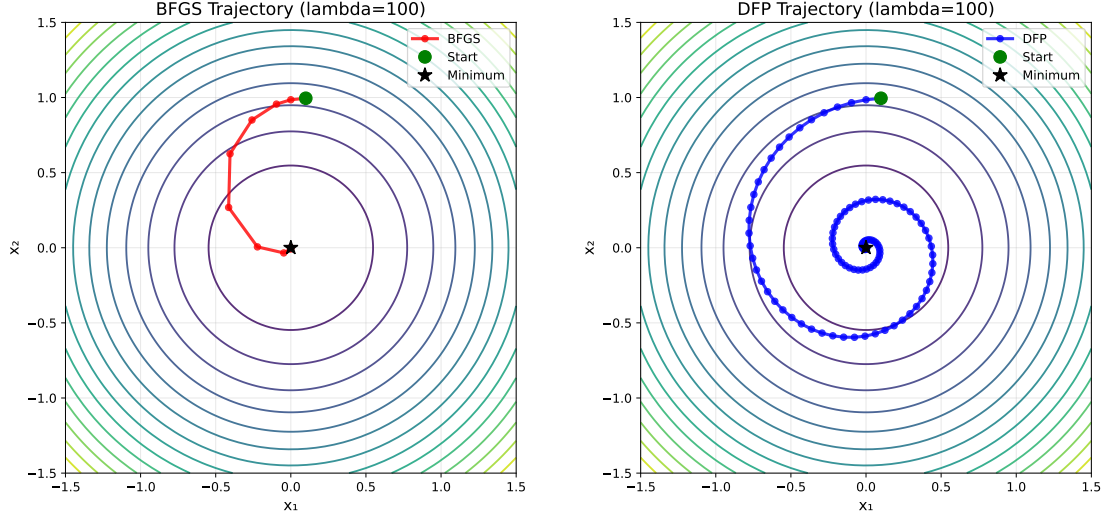


Figure 3: Iterative trajectories for BFGS and DFP methods when $\lambda_1 = 100$. BFGS converges in 10 iterations while DFP requires 107 iterations, showing BFGS's superior efficiency for large eigenvalue errors.

Practical Implications These findings provide strong empirical justification for the widespread practical preference for BFGS over DFP. The analysis of Powell's simple quadratic problem yields deep insights about quasi-Newton methods' behavior far from the solution, where most computational effort is expended. The superior efficiency of BFGS in correcting erroneous Hessian approximations makes it the algorithm of choice for robust and efficient unconstrained optimization.

1.5 Limited Memory BFGS (L-BFGS)

Limited-memory quasi-Newton methods extend classical quasi-Newton methods to large-scale optimization problems. In standard quasi-Newton methods, the approximate Hessian or inverse Hessian matrix is stored and updated as a dense matrix, requiring $\mathcal{O}(n^2)$ memory for n variables.

The L-BFGS method [6], which is based on the BFGS update, avoids storing the full matrix explicitly. Instead, it maintains only the most recent m vector pairs $\{(s_i, y_i)\}$. This reduces the storage requirement to $\mathcal{O}(nm)$, which is a dramatic improvement when m is a small constant (typically $m \leq 10$).

Throughout this subsection, we work with a finite sequence of matrices

$$H_0, H_1, \dots, H_m,$$

where H_ℓ denotes the inverse Hessian approximation obtained after ℓ BFGS updates

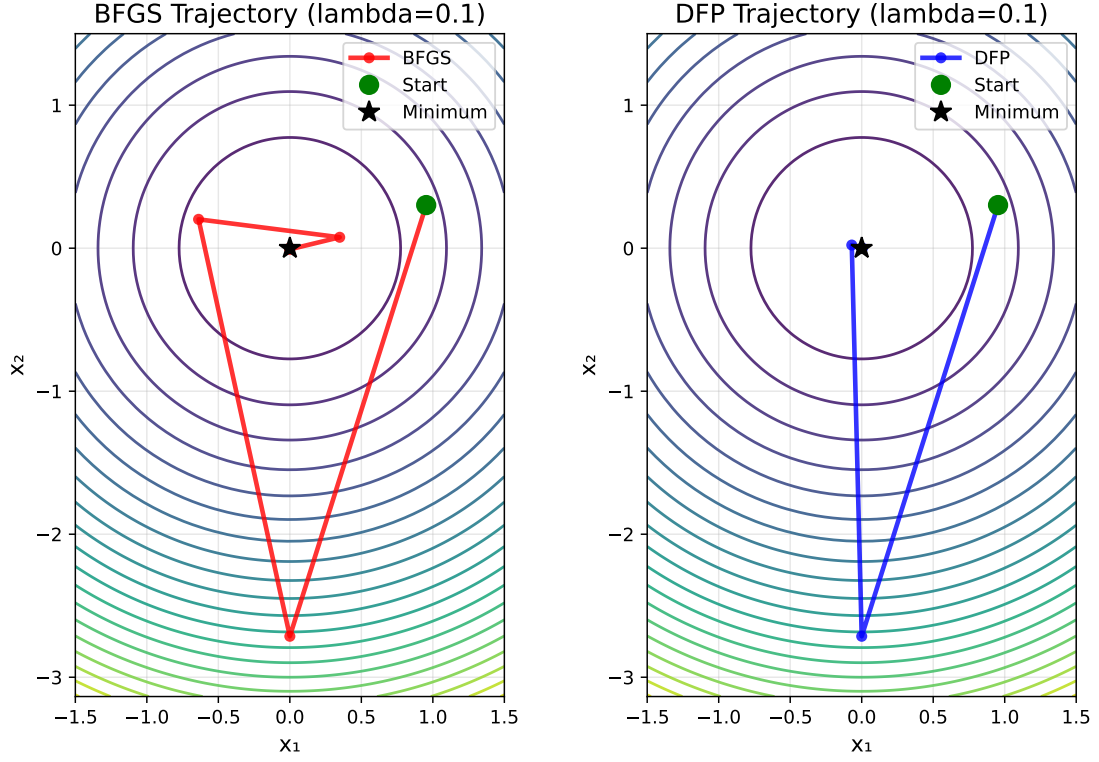


Figure 4: Iterative trajectories for BFGS and DFP methods when $\lambda_1 = 0.1$. Both methods converge quickly, with DFP slightly faster than BFGS, demonstrating the symmetric behavior when eigenvalues are underestimated.

applied to a given initial matrix H_0 . Note that this is not the same as the iterates in an optimization algorithm. Here, we focus solely on the structure of the BFGS updates.

1.5.1 Compact Representation of H_m

Let $\{(s_i, y_i)\}_{i=0}^{m-1}$ be the stored correction pairs, and define

$$\rho_i = \frac{1}{y_i^\top s_i}, \quad V_i = I - \rho_i y_i s_i^\top. \quad (58)$$

From eq. (46), the BFGS update for the inverse Hessian can be expressed as

$$H_{i+1} = V_i^\top H_i V_i + \rho_i s_i s_i^\top, \quad (59)$$

Algorithm 1: L-BFGS Two-Loop Recursion for $r = H_m q$ [7, Algorithm 7.4]

Input: q , stored pairs $\{(s_i, y_i)\}_{i=0}^{m-1}$, initial matrix H_0
Output: $r = H_m q$

```

1 for  $i = m - 1, m - 2, \dots, 0$  do
2    $\rho_i \leftarrow 1 / (y_i^\top s_i)$ 
3    $\alpha_i \leftarrow \rho_i s_i^\top q$ 
4    $q \leftarrow q - \alpha_i y_i$ 
5 end
6  $r \leftarrow H_0 q$ 
7 for  $i = 0, \dots, m - 1$  do
8    $\beta_i \leftarrow \rho_i y_i^\top r$ 
9    $r \leftarrow r + s_i(\alpha_i - \beta_i)$ 
10 end
11 return  $r$ 

```

for $i = 0, \dots, m - 1$. By recursively expanding this relation, we obtain the compact representation

$$H_m = V_{m-1}^\top \cdots V_0^\top H_0 V_0 \cdots V_{m-1} + \sum_{j=0}^{m-1} (V_{m-1}^\top \cdots V_{j+1}^\top) \rho_j s_j s_j^\top (V_{j+1} \cdots V_{m-1}), \quad (60)$$

where H_0 denotes the chosen initial inverse Hessian approximation, typically a scaled identity matrix.

1.5.2 Two-Loop Recursion

The compact representation above can be applied to any vector q without explicitly forming H_m . Let $r = H_m q$. Exploiting the associative structure of the matrix products, this operation can be carried out using two short loops of length m , leading to the well-known L-BFGS two-loop recursion [7, Algorithm 7.4]. The pseudocode is provided in Algorithm 1. This algorithm requires $\mathcal{O}(md)$ arithmetic operations and $\mathcal{O}(md)$ storage, where d denotes the problem dimension.

Now, we verify that the output of Algorithm 1 indeed computes $r = H_m q$.

Proposition 8. *The output of the two-loop recursion Algorithm 1 satisfies $r = H_m q$.*

Proof. In the backward recursion, for $i = m - 1, m - 2, \dots, 0$, from the input vector $q^{(m)} := q$, the algorithm computes

$$\alpha_i = \rho_i s_i^\top q^{(i+1)}, \quad q^{(i)} := q^{(i+1)} - \alpha_i y_i. \quad (61)$$

Substituting the definition of α_i and using eq. (58), we find

$$q^{(i)} = q^{(i+1)} - \rho_i \left(s_i^\top q^{(i+1)} \right) y_i = \left(I - \rho_i y_i s_i^\top \right) q^{(i+1)} = V_i q^{(i+1)}.$$

Thus, for all $i = 0, 1, \dots, m-1$, we have

$$q^{(i)} = V_i V_{i+1} \cdots V_{m-1} q.$$

Then, the algorithm applies the initial inverse Hessian approximation:

$$r^{(0)} = H_0 q^{(0)} = H_0 V_0 V_1 \cdots V_{m-1} q.$$

Next, for $i = 0, 1, \dots, m-1$, the forward recursion computes

$$\beta_i = \rho_i y_i^\top r^{(i)}, \quad r^{(i+1)} = r^{(i)} + s_i (\alpha_i - \beta_i).$$

Substituting the definitions of α_i , β_i , and $q^{(i+1)}$, we have

$$\begin{aligned} r^{(i+1)} &= r^{(i)} + \rho_i s_i s_i^\top (V_{i+1} V_{i+2} \cdots V_{m-1}) q - \rho_i s_i y_i^\top r^{(i)} \\ &= \left(I - \rho_i y_i s_i^\top \right) r^{(i)} + \rho_i s_i s_i^\top (V_{i+1} V_{i+2} \cdots V_{m-1}) q \\ &= V_i^\top r^{(i)} + \rho_i s_i s_i^\top (V_{i+1} V_{i+2} \cdots V_{m-1}) q. \end{aligned}$$

By recursively expanding this relation from the initial value $r^{(0)} = H_0 q^{(0)}$, we obtain

$$r^{(m)} = V_{m-1}^\top \cdots V_0^\top H_0 V_0 \cdots V_{m-1} q + \sum_{j=0}^{m-1} (V_{m-1}^\top \cdots V_{j+1}^\top) \rho_j s_j s_j^\top (V_{j+1} \cdots V_{m-1}) q, \quad (62)$$

which matches the compact representation of H_m in eq. (60) applied to q , completing the proof. $\square$

Thus, the two-loop recursion exactly evaluates the action of H_m on a vector while avoiding explicit matrix construction, achieving both mathematical rigor and computational efficiency.

1.5.3 Initial Step Size

A crucial component of the L-BFGS method is the choice of the initial matrix H_0 . A widely used and well-justified option is a scaled identity,

$$H_0 = \gamma I, \quad (63)$$

where the scaling parameter is chosen as

$$\gamma = \frac{s_{m-1}^\top y_{m-1}}{y_{m-1}^\top y_{m-1}}. \quad (64)$$

This choice is motivated by the relationship between the approximate inverse Hessian and the local curvature of the objective function [6, 9]. To justify this scaling, assume that the objective function f is twice continuously differentiable and consider the mean Hessian along the most recent step:

$$\bar{G} = \int_0^1 \nabla^2 f(x + \tau s_{m-1}) d\tau, \quad (65)$$

where s_{m-1} denotes the most recent displacement. By the mean value theorem,

$$y_{m-1} = \nabla f(x + s_{m-1}) - \nabla f(x) = \bar{G} s_{m-1}.$$

Using this relation, the scaling factor can be written as

$$\frac{s_{m-1}^\top y_{m-1}}{y_{m-1}^\top y_{m-1}} = \frac{(\bar{G}^{1/2} s_{m-1})^\top (\bar{G}^{1/2} s_{m-1})}{(\bar{G}^{1/2} s_{m-1})^\top \bar{G} (\bar{G}^{1/2} s_{m-1})}, \quad (66)$$

which is a Rayleigh quotient. If $\bar{G}^{1/2} s_{m-1}$ happens to be an eigenvector of $\bar{G}$, then the reciprocal of this quantity equals the corresponding eigenvalue.

Moreover, the choice (64) coincides with the short step size of the Barzilai–Borwein method [1], highlighting a close connection between L-BFGS initialization and classical step-length selection strategies. This observation further supports the effectiveness of the scaled-identity initialization in practice.

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